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Cataract Update

Hereditary pediatric cataract on the Arabian Peninsula

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Abstract

Hereditary pediatric cataract on the Arabian Peninsula does not follow the same epidemiological patterns as described for Western populations. This article describes selected genetic causes for inherited pediatric cataract in the region.

Keywords: Pediatric cataract, Genetics, Arabian Peninsula, Crystallins

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Introduction

Non-traumatic pediatric cataract is uncommon. Unilateral cases are often idiopathic and typically occur in the smaller eye with persistent fetal vasculature that can range from subtle to obvious. Bilateral cases are more likely to have a systemic association or a detectable cause such as malnutrition, metabolic imbalance, infection, drug exposure, or gene mutation. For those cases of apparently non-syndromic congenital or childhood-onset (juvenile) cataract caused by gene mutations, the most obvious candidates are genes responsible for normal lens structure such as lens structural proteins, lens cytoskeletal proteins, certain collagens, and/or ocular transcription factors. Many such gene mutations have been described, particularly for genes that encode the crystallins, which are the major water-soluble structural component of the lens.[1–3] In general the ability to make phenotype-genotype correlations in hereditary pediatric cataract is limited because of variable expressivity (phenotypes) for a given genotype and overlapping phenotypes for different genotypes; however, there are exceptions.

In Arab countries, epidemiological and genetic data regarding hereditary pediatric cataract are lacking. In the West, studies have shown that up to one-third of the bilateral cases are genetic, most of which are autosomal dominant.[2–4] However, these statistics are not applicable to Saudi Arabia or to the surrounding countries.[5] While dominant causes occur, the social preference for intrafamilial marriage and a large number of children is expected to increase expression of potential homozygous recessive disease and thus the incidence of both known and novel recessive causes for hereditary cataract. This has been my experience from genetic counseling of affected Saudi families. The current report illustrates selected genetic causes for hereditary pediatric cataract on the Arabian Peninsula.

Lens structure (crystallins)

Crystallins comprise the major structural component of the human lens.[1] Thus it is not surprising that mutations in the crystallin genes are estimated to account for approximately half of non-syndromic hereditary pediatric cataract.[3] Three major mammalian crystallin groups are distinguishable based on size, charge and immunological properties: alpha (α), beta (β), and gamma (γ).[1]

Most reported crystallin mutations are dominant[1–3]; however, in my experience, most Saudi families with hereditary

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cataract tend to harbor recessive crystallin mutations. This is often in one of the β-crystallins (unpublished data).

Selected example 1 – α-crystallin (CRYAA and CRYAB)

α-crystallin, the major protein component of the mammalian lens, is an aggregate comprised of 2 gene products: alpha-A (from the gene *CRYAA*) and alpha-B (from the gene *CRYAB*).[1] α-crystallin facilitates the correct folding of other lenticular proteins *in vivo*, and thus it is important in suppressing protein aggregation and maintaining lens transparency.

CRYAA encodes the alpha-A component of α-crystallin and is expressed mainly in the lens.[1] Most reported mutations are dominant.[2] One of the two reported recessive *CRYAA* mutations (p.R54C) caused congenital white cataract in a consanguineous nuclear family from the southern Arabian Peninsula with 3 out of 10 children affected.[6] Affected children also had microcornea, which may have been from abnormal cornea induction by the abnormally-formed lens (Fig. 1). In that family asymptomatic mutation carriers were documented to have asymptomatic lenticular opacities, a clinical heterozygote carrier sign that had not been previously documented for recessive cataract.[6]

CRYAB encodes the alpha-B component of α-crystallin. Unlike *CRYAA*, *CRYAB* has a relatively broad expression pattern that includes muscle and retina, where it serves a protective anti-apoptotic function.[1] Virtually all reported mutations are dominant, with phenotypes that can include myopathy with or without cataract. A recessive p.R56W CRYAB mutation was documented in a Saudi family from the eastern Arabian Peninsula.[7,8] In that family, of the 12 studied family members 7 were homozygous for the mutation. These homozygous individuals had lens phenotypes that ranged from significant congenital cataract to visually-insignificant lens opacities during childhood or adulthood (Fig. 2). This was significant because it demonstrated variable expressivity for the same pathological genotype, a phenomenon that had been well-documented for dominant cataract[9] but not for recessive cataract. Variable expressivity for familial cataract is related to the influence of other background genetic factors and/or environment. Interestingly, those family members homozygous for the *CRYAB* mutation who were left aphakic after childhood cataract surgery developed retinal degeneration decades later while those individuals homozygous for the mutation who were made pseudophakic or left phakic did not.[7,8] This was possibly from impaired anti-apoptotic CRYAB function in the retina and highlights how environment (unfiltered light in an aphakia as opposed to pseudophakia or phakia) can potentially interact with underlying genotype (the *CRYAB* mutation as expressed in the retina) to cause phenotypic consequences (Fig. 3).



Figure 1. This child with recessive *CRYAA* mutation (p.R54C) had infantile cataract surgery and was fitted with aphakic glasses. The associated microcornea can be appreciated.

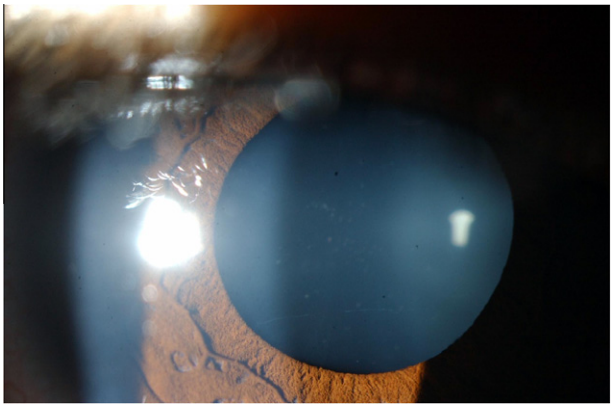


Figure 2. Although some of her affected relatives with recessive *CRYAB* mutation (p.R56W) had total white cataract within the first few years of life, the ocular phenotype of this girl harboring the recessive mutation consists of asymptomatic punctate lens opacities. This demonstrates variable expressivity for a cataract-causing recessive mutation.

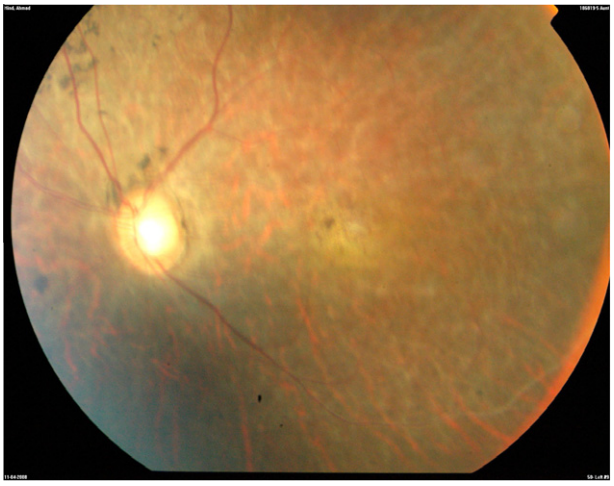


Figure 3. This woman had cataract surgery in early childhood for *CRYAB*-related cataract (recessive p.R56W mutation) and was left aphakic. Decades later she developed retinal degeneration. This was presumably related to disruption of the normal anti-apoptotic function of retinal CRYAB in the setting of many years of unfiltered light reaching the retina.

Selected example 2 – γ-D-crystallin (CRYGD)

Both γ-crystallins and β-crystallins are comprised of homologous Greek key motifs with 2 domains. The major differences are that γ-crystallins are monomeric and lack terminal extensions. Of the γ-crystallins, only *CRYGC* and *CRYGD* are significantly expressed in the human lens.[10] A specific founder dominant mutation in *CRYGD* (p.P23T) was documented as the cause for a unique phenotype of juvenile cerulean cataract in Saudi children from 2 different families (Fig. 4).[11] Haplotype analysis confirmed that the mutation was from a single founder. One affected sibling who harbored the mutation had coralliform cataract, suggesting that coralliform and cerulean cataract represent variable expressivity for the genotype rather than distinct conditions.[11] Inter-

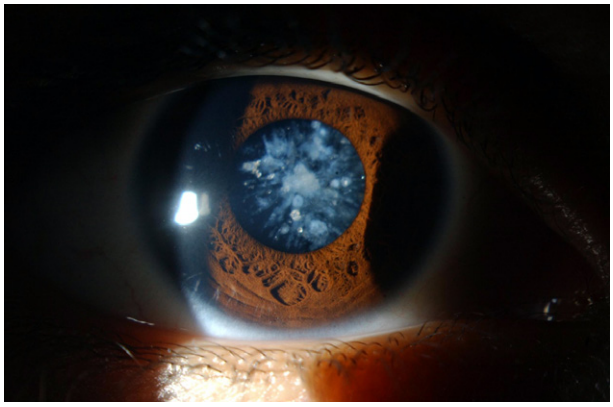


Figure 4. The classic cerulean cataract phenotype is shown in a Saudi child. The cause in this child, dominant p.P23T CRYGD mutation, was also found to underly the phenotype in another Saudi family. These 2 Saudi families were not known to be related but haplotype analysis suggested they inherited the mutation from a common ancestor.

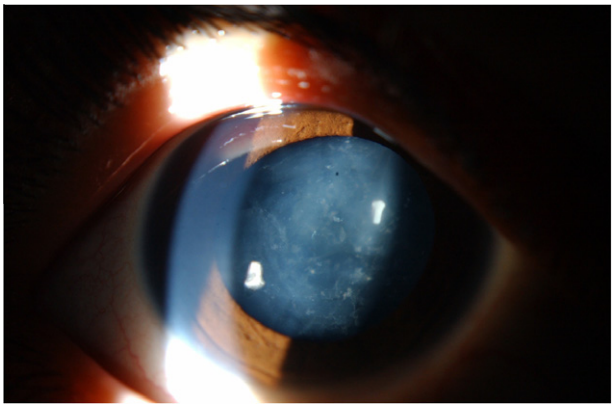


Figure 5. At approximately 12 years of age, this girl and her affected sisters became symptomatic from juvenile cataract due to recessive *BPSP2* mutation (p.A201R*fs*X19). This cataract is from impairment in a cytoskeletal protein and its morphology is unique – diffuse cortical cataract with scattered lens opacities.

estingly, the p.P23T CRYGD mutation was also previously associated with cerulean or coralliform cataract once before in a Moroccan family,[12,13] raising the possibility that the affected individuals in the Saudi families and the Moroccan family inherited the mutation from a common ancestor.[11]

Lens cytoskeletal proteins

Lens cytoskeletal proteins are important in optimal lens fiber cell organization such that the crystalline lens maintains its optical properties. Mutations in these genes can give unique phenotypes.

Selected example 1 – beaded filament structural protein 2 (BFSP2)

BFSP2 is an intermediate filament protein highly expressed in the lens and critical to proper lens fiber architecture.[14] Dominant cataract-causing mutations have been described.[14] The one documented recessive *BFSP2* mutation was in a Saudi family from the northern Arabian Peninsula (p.A201RfsX19).[15] Three sisters out of 9 siblings developed a unique phenotype of juvenile-onset diffuse cortical cataract with scattered fleck lens opacities at approximately 12 years of age (Fig. 5).[15] The carrier mother did not have significant lens opacities while the carrier father had had surgery for posterior subcapsular cataract in his mid-forties, which seems to have been coincidental. The recessive *BFSP2* mutation in this family did not seem to have manifestations in the carrier (heterozygous) state and thus likely lacked the ability to assemble, polymerize, or interfere with the normal protein when in the carrier state. This was unlike the previously-reported mutations in this gene, which were dominant and presumably interfered with normal lens fiber maintenance.[15]

Selected example 2 – NHS

The gene *NHS* encodes a regulator of actin remodeling and cell morphology and is particularly expressed in the lens, brain, craniofacial mesenchyme, and dental primordia.[16] Mutations in *NHS* cause the Nance-Horan syndrome, an often under-diagnosed X-linked developmental disorder characterized by congenital or infantile cataract with microcornea in boys.[16] Non-ophthalmic features include particular facial features (long face, bulbous nose, rotated ears, dental abnormalities), variable mental retardation (up to 30% of affected boys), and shortened lateral metacarpals.[17] For ophthalmic X-linked disease in general, females from affected families can be asymptomatic despite harboring the gene mutation and can potentially be clinically identified by certain carrier signs. The study of a large affected Saudi family harboring *NHS* mutation from the western Arabian Peninsula has provided insight into female carrier signs for Nance-Horan syndrome (*Ophthalmic Genetics*, in press). The 4 genetically-confirmed asymptomatic female carriers of a specific *NHS* mutation could be identified clinically by their characteristic Y-centered lens opacities (Fig. 6) in addition to their characteristic facial features. No potential female carrier without these findings from the family was a carrier for the *NHS* mutation, not even a potential female carrier with punctate lens opacities (that were not Y-centered).

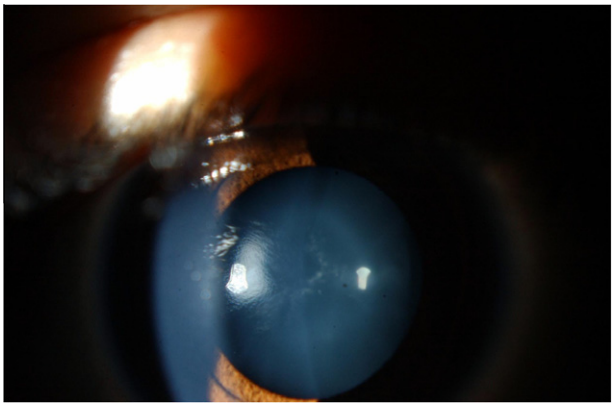


Figure 6. In this asymptomatic female harboring an *NHS* mutation, the Y-centered lens opacities are a clinical carrier sign.

Certain collagens

Collagens are a heterogeneous group of extracellular glycoproteins with a triple helical domain motif that have different structural and matrix functions throughout the body.

Thus mutations in collagen-encoding genes often have protean manifestations. In some instances, they result in characteristic forms of congenital or juvenile cataract that if recognized can be helpful in the diagnosis of certain syndromes.

Selected example 1 – collagen type II alpha 1 (COL2A1, Stickler syndrome)

Sticker syndrome is a clinical diagnosis with variable features but characterized by midface hypoplasia, hearing loss, spondyloepiphyseal dysplasia, and ophthalmic findings. Ophthalmic findings include high myopia, vitreoretinal degeneration, a propensity for retinal detachments, and a characteristic wedge-shaped partial cataract. Although cataracts in Stickler syndrome do not always have this specific morphology, when it is present it should raise suspicion for the diagnosis (Fig. 7). The etiology for this particular cataract morphology is unclear. Stickler syndrome is genetically heterogeneous, although many cases seem to be caused by dominant mutation in *COL2A1*, a fibrillar collagen highly expressed in the vitreous and cartilage.[18] Interestingly, patients with mutation in exon 2 of *COL2A1* have a predominantly ocular phenotype with minimal extraocular features.[19] The findings of partial cataract and vitreous degeneration (Fig. 8) in a child should raise suspicion for ocular Stickler syndrome, especially in the setting of high myopia. While dominant *COL2A1* mutations cause the cataract associated with Stickler syndrome in Saudi Arabia as they do in the rest of the world, in this region the phenotype is also caused by recessive mutations in other collagen genes (unpublished data).

Selected example 2 – collagen type XVIII alpha-1 (COL18A1, Knobloch syndrome)

Collagen XVIII alpha-1 is one of the multiplexins, extracellular matrix proteins that contain multiple triple-helix domains interrupted by non-collagenous domains.[20] The proteolytically produced C-terminal fragment of type XVIII collagen is endostatin, a potent antiangiogenic protein. The short isoform of collagen XVIII is expressed in brain, retina, and vascular and epithelial basement membranes throughout the body.[20,21] Recessive mutations in *COL18A1* cause

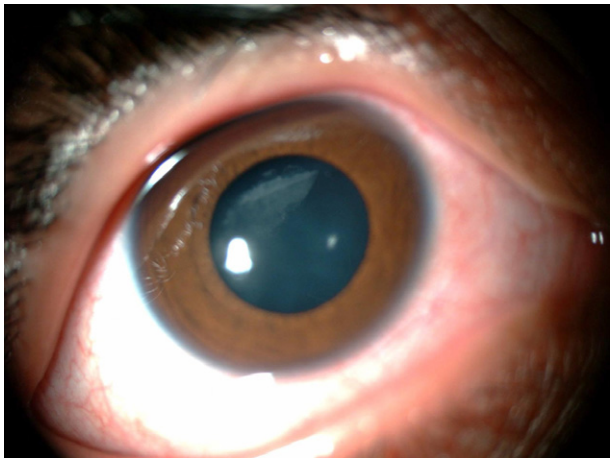


Figure 7. In Stickler syndrome, cortical wedge-shaped partial cataract is a classic finding. Although cataracts in Sticker syndrome do not always have this morphology, when it is present it is helpful in diagnosing the syndrome.

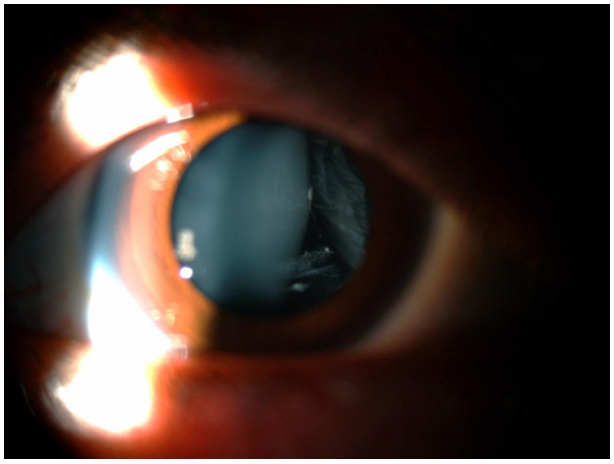


Figure 8. In addition to the cortical wedge-shaped partial cataract, vitreous degeneration immediately posterior to the lens is another helpful sign in diagnosing Stickler syndrome.

Knobloch syndrome, characterized by occipital defect, high myopia, vitreoretinal degeneration, and a propensity for retinal detachments.[20,22] Ectopia lentis also often occurs. Affected children tend to develop a characteristic peripheral crescentic lens opacity that when present is helpful in confirming the diagnosis. In fact, when present with other ocular features that characterize the syndrome, in my experience the ocular phenotype becomes pathognomonic for the diagnosis of Knobloch syndrome. The etiology of this particular lens morphology is not known.

Ocular transcription factors

Normal anterior segment development is dependent upon several genes that orchestrate and regulate the formation of the eye, examples of which include but are not limited to *PAX6*, *FOXC1*, and *PITX2*. Dominant mutation in one of these transcription factors is a potential cause of hereditary cataract, often in the context of other ocular developmental disease and sometimes also in the context of non-ocular findings. Phenotypic variability and overlap from mutations in these genes often occur.

Selected example: paired box gene 6 (PAX6)

PAX6 is a transcription factor with an instrumental role in orchestrating ocular development.[23] Recessive *PAX6* mutations are lethal.[24] Dominant *PAX6* mutations cause aniridia, a pan-ocular disorder that can affect the conjunctiva, cornea, lens, iris, trabecular meshwork, fovea, and/or optic nerve. The classic anirida phenotype is specific for dominant *PAX6* mutation although infrequently patients with the classic aniridia phenotype do not have detectable *PAX6* mutation.[23,25] When an insult that causes *PAX6* mutation also disrupts the neighboring gene *WT1*, which is more likely in sporadic cases, affected children have a predisposition to Wilms tumor of the kidney.[26]

PAX6 mutations do not always cause an obvious aniridic phenotype.[23] Variant phenotypes can occur, particularly with missense mutations.[27] In fact, *PAX6* mutations can cause seemingly-isolated congenital or juvenile cataract.[24,27] The morphology of *PAX6*-related cataract varies, but anterior

polar lens opacities are common. As is true for generally for dominant disease, on the Arabian Peninsula *PAX6*-related phenotypes are not expected to be more or less frequent than in the rest of the world.

Summary

There are many potential genetic etiologies for hereditary pediatric cataract. In the West, familial pediatric cataract is often dominant. While dominant mutations are a potential cause on the Arabian Peninsula, in this region of the world recessive causes comprise a larger percentage of cases and some of these causes are unique. Additional studies regarding the epidemiology and genetics of hereditary pediatric cataract in the region are needed, both to further understanding of crystalline lens pathophysiology and also to improve local genetic counseling services.

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